

EDUCATION

Nguyen Khanh An

VinUniversity*B.S. in Data Science, Minor in Economics*

Hanoi, Vietnam

Sep 2024 - Present

- **Relevant Coursework:** Calculus, Linear Algebra, Statistics, Econometrics, Object-Oriented Programming, Data Structures, Microeconomics, Macroeconomics

EXPERIENCE

Swartz Center for Computational Neuroscience (SCCN), UC San Diego*Visiting Research Student*

La Jolla, CA

Expected Jun 2026 -

- Incoming research supervised by Prof. Tzyy-Ping Jung at SCCN - the lab behind EEGLAB and seminal work on Independent Component Analysis for EEG / brain-computer interfaces.

LSM's CoLab, VinUniversity/Lancaster University*Undergraduate Research Assistant*

Hanoi, Vietnam

Feb 2026 - Present

- Expanded testing below the original study's scope into highly non-IID data distributions (Dirichlet $\alpha = 0.1$), uncovering three novel boundary behaviors: defense success at 20% malicious clients, severe instability at 40%, and a complete federated-training collapse at 60%.
- Quantified model vulnerabilities using Wasserstein distance and percentile-band overlap for score-distribution analysis; provided new diagnostic evidence absent from the original paper demonstrating a $5\times$ drop in benign/malicious separability ($W = 1.258 \rightarrow 0.243$) across the empirical failure boundary.

College of Arts and Sciences, VinUniversity*Tutor - Statistics for Social Sciences*

Hanoi, Vietnam

Jan 2026 - Present

- Supported instruction for STAT1011 (Statistics for Social Sciences, Spring 2026), facilitating review sessions and holding recitation classes for undergraduate students.

VinUniversity Library*Technical Media Developer*

Hanoi, Vietnam

Sep 2024 - Jan 2026

- Create augmented reality (AR) learning experiences by designing 3D assets in Blender and deploying them via online AR platforms.
- Develop and optimize 3D models for AR/VR use (topology, UV mapping, texturing, and real-time performance constraints).

PROJECTS

VinUni Room Booking*Frontend Lead (Team of 4)*

Hanoi, Vietnam

Apr 2026 - May 2026

- Built the entire frontend on Vaadin Flow + Spring Boot, persisting to MySQL via JDBC.
- Designed an invite-only booking flow (multi-select attendee picker, SQL-based conflict detection, six validator rules covering access, duration, advance window, minimum participation, one-per-day, past-time) and an admin approval queue with auto-approval policy by room category.
- Led an end-to-end product audit identifying 21 bugs across model/service/UI layers; shipped a single PR removing ~ 500 lines of dead code, extending test mocks, and closing every audit finding.

Weather DashBoard — EcoBox*Facilitator*

Hanoi, Vietnam

Dec 2025 - Jan 2026

- Engineered the physical enclosure using AutoCAD, calculating precise tolerances to house the 7.5-inch E-Ink HAT and PCB before 3D printing the final assembly in PLA plastic.
- Helped implement a hybrid rendering pipeline using Python and headless browser automation to convert dynamic HTML/CSS interfaces into static bitmaps, optimizing color mapping for the limited 3-color (Black, White, Yellow) E-ink palette.

HONORS & AWARDS

- Dean’s List, Spring 2025:

SKILLS

Programming Languages: Python, R, L^AT_EX